COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	07/15/26	**DATE SUBMITTED:**	07/15/26
NAME:	Hare, Andrei Angelo D.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. ___ Seatwork 2.1: Memory Diagram Exercise

1. PROCEDURE OUTPUT

What procedures did you perform?

> we answer the question related on creating a memory diagram from the given values

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini, Copilot**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: *idea generation, grammar correction*

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I only research the key concepts on how the questions work and create on my own based on my understanding but still unsure about my answers.**

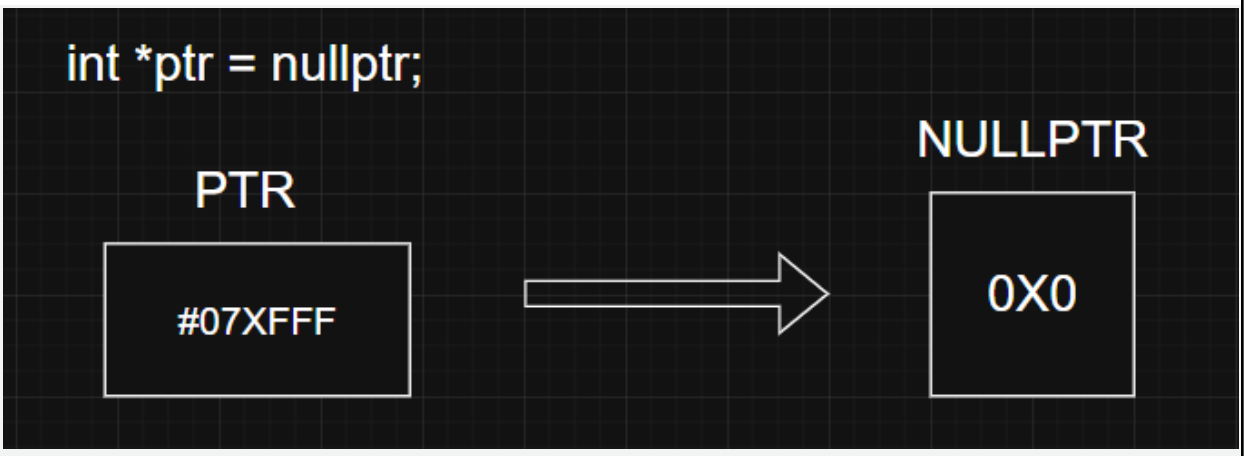
Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

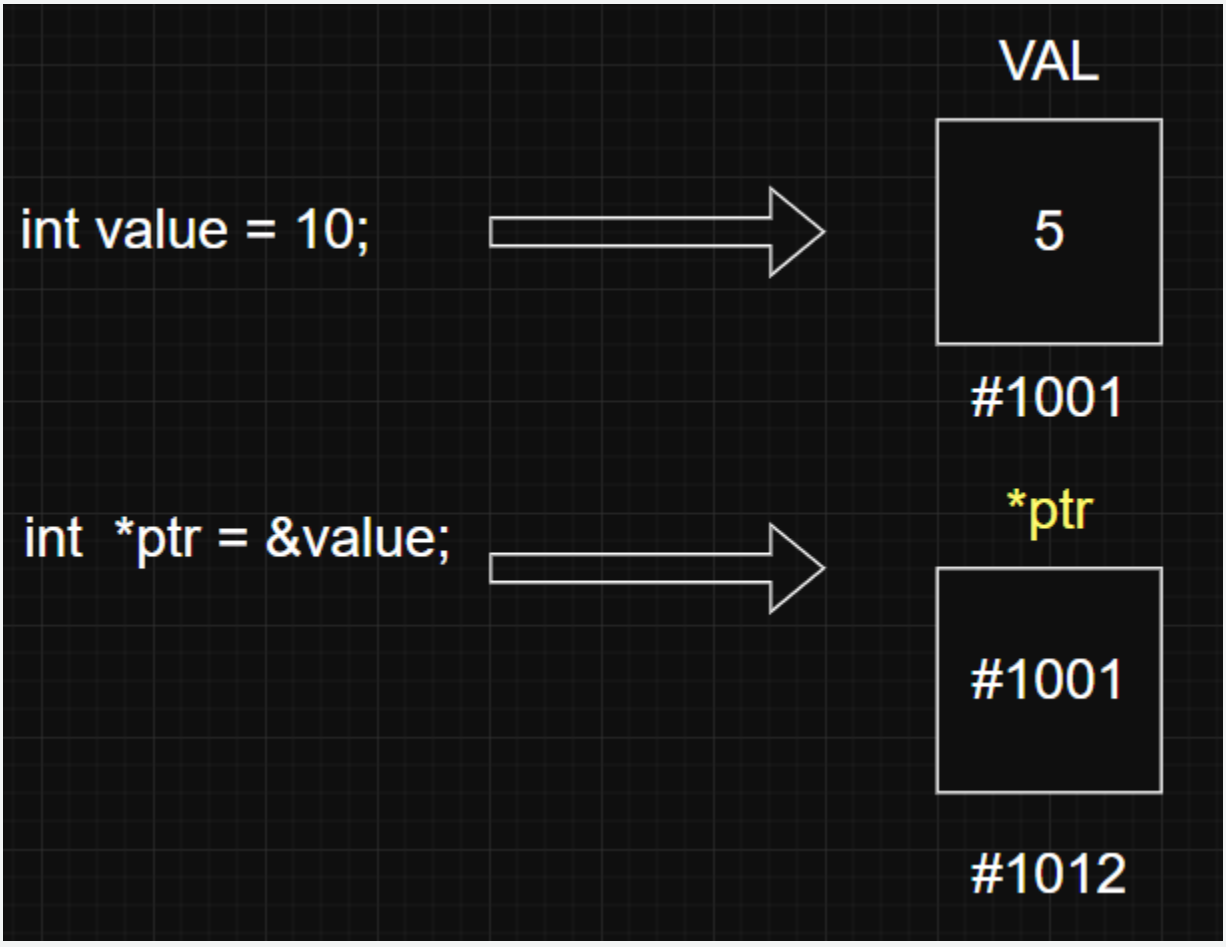
I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Draw the memory diagram for the following:

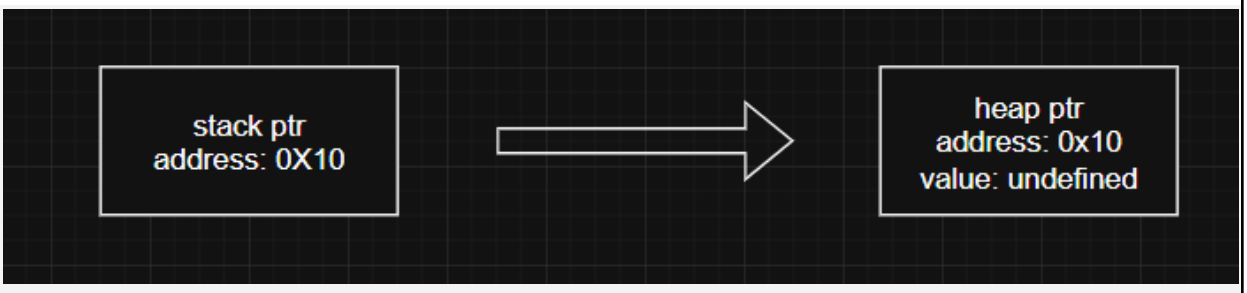
```
1. int *ptr = nullptr;
```



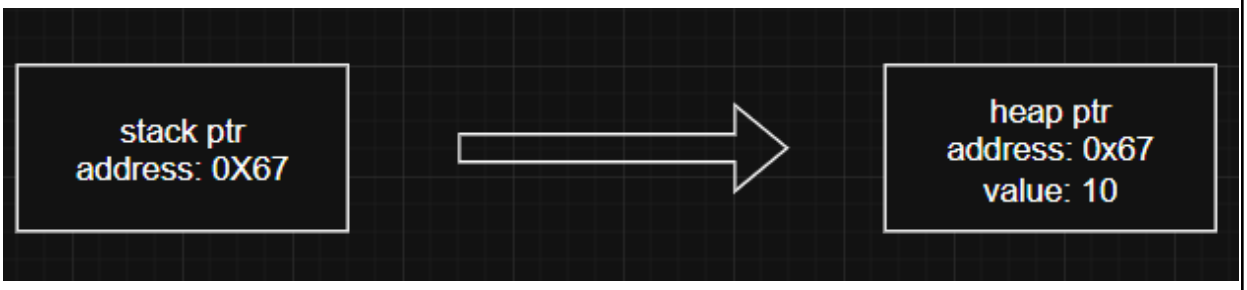

```
2. int value = 10;
   int  *ptr= &value;
```




```
3. int *ptr = new int;
```



```
4. int *ptr = new int(10);
```



5. Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Operator.

The constructor **brings the object to life**, while the destructor **safely deletes / releases it**. The copy constructor **builds a new object from scratch**. The assignment operator takes a **pre-existing object and updates its contents**.

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